JPX ENSEMBLE | FULL EIGHT-FOLD STUDY

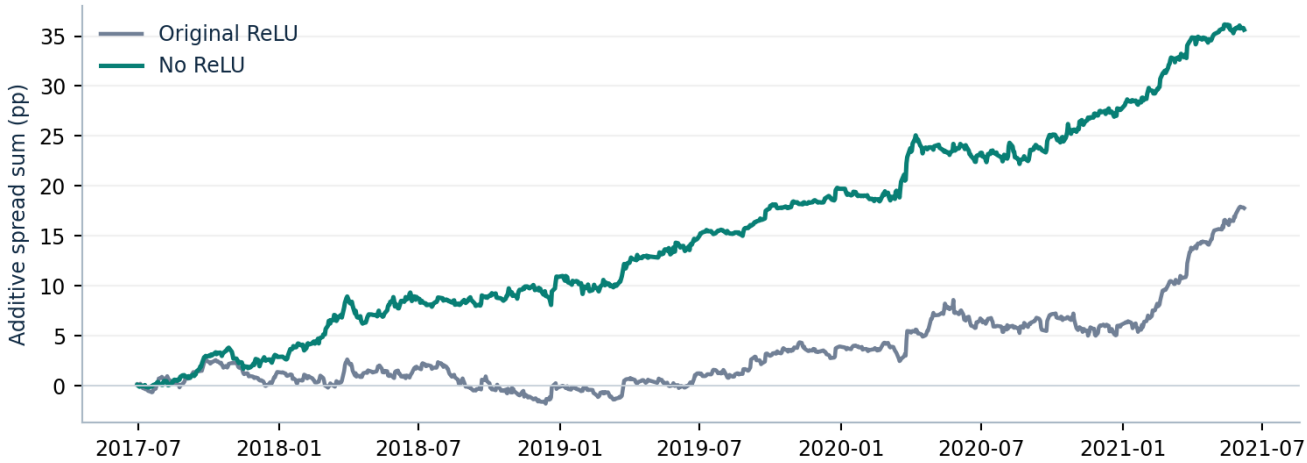
Does removing ReLU help?

Completed 19 October 2026, 19:28 Taipei | 160 epochs | 8 chronological validation folds

On exactly matched held-out rows, removing the dense ReLU increased pooled gross Sharpe from **0.07678** to **0.14462**. MAE was 0.97% higher (worse). This is a one-seed historical comparison; portfolio ranking and numerical forecast accuracy must be assessed separately.

Coverage: 1,853,400 security-date predictions across 958 dates, 2017-06-30 to 2021-06-08. No untouched final test period was reserved.

Pooled out-of-sample measure	Original ReLU	No ReLU
Sharpe, unannualized	0.07678	0.14462
MAE	0.01606012	0.01621575
Huber loss, delta = 1	0.00029055	0.00029401
Zero-forecast MAE	0.01597436	0.01597436
Mean daily Rank IC, native defined dates	0.01899	0.00508
Rank IC defined dates	548	957
Gross additive spread sum	17.78 pp	35.64 pp



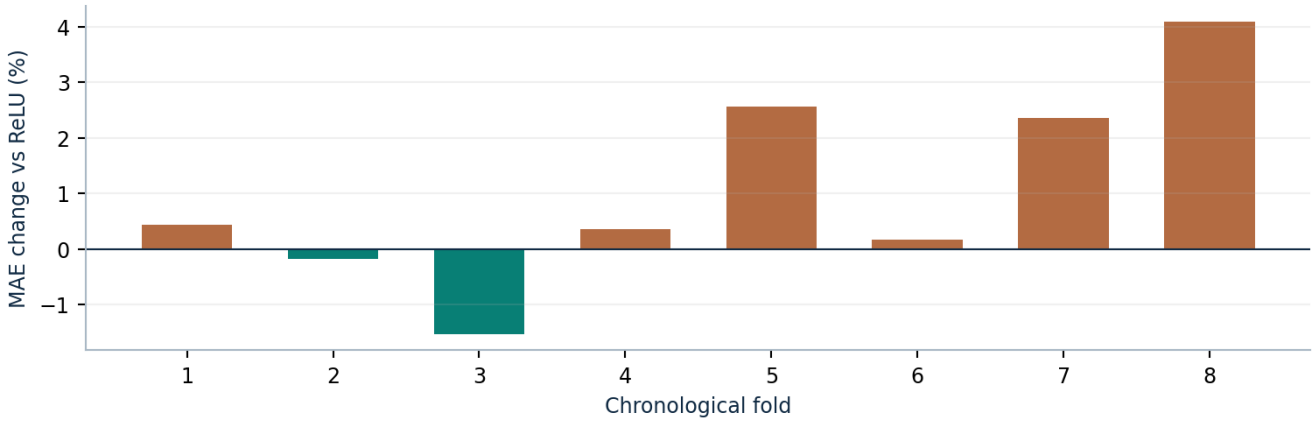
The chart adds daily spreads. Percentage-point units do not represent a compounded account return. All primary Sharpe values are unannualized; costs are excluded here.

PREDICTION ERRORS

Accuracy by validation fold

Each row compares the same securities, dates, float32 Targets and preprocessing context. MAE and Huber are calculated from validation predictions; they are not training-loss substitutes.

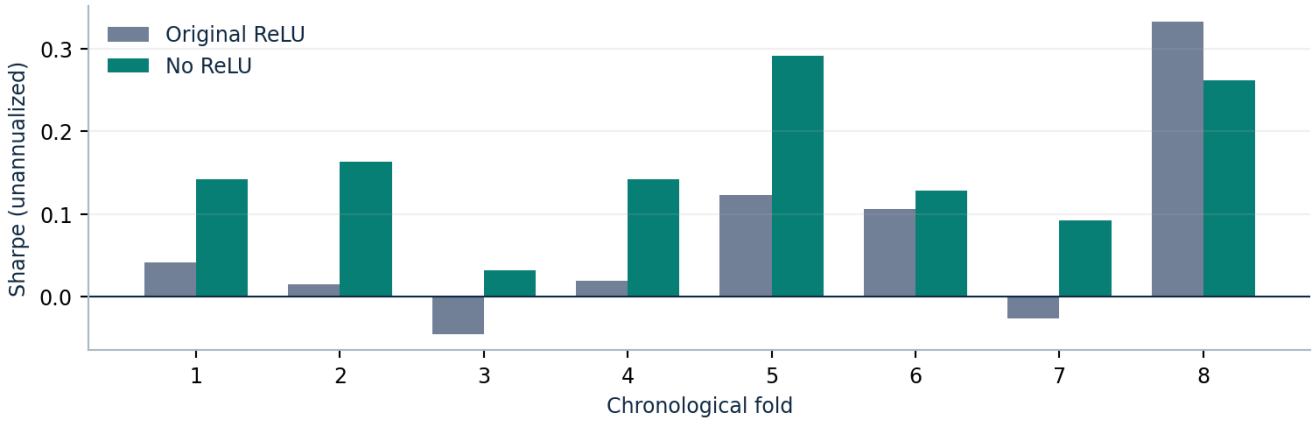
Fold / dates / samples	MAE ReLU	MAE No ReLU	Zero MAE	Huber ReLU	Huber No ReLU
Fold 1 | 122 dates 2017-06-30 to 2017-12-26 228,969 samples	0.012171	0.012224	0.012123	0.0001845	0.0001852
Fold 2 | 119 dates 2017-12-27 to 2018-06-22 225,375 samples	0.014975	0.014948	0.014707	0.0002564	0.0002560
Fold 3 | 125 dates 2018-06-25 to 2018-12-21 238,566 samples	0.017460	0.017193	0.017017	0.0003233	0.0003173
Fold 4 | 114 dates 2018-12-25 to 2019-06-19 219,599 samples	0.015815	0.015873	0.015819	0.0002794	0.0002815
Fold 5 | 121 dates 2019-06-20 to 2019-12-16 234,905 samples	0.013341	0.013683	0.013376	0.0001977	0.0002038
Fold 6 | 117 dates 2019-12-17 to 2020-06-12 229,702 samples	0.022400	0.022438	0.022472	0.0005305	0.0005316
Fold 7 | 121 dates 2020-06-15 to 2020-12-10 239,011 samples	0.016934	0.017335	0.016914	0.0003077	0.0003184
Fold 8 | 119 dates 2020-12-11 to 2021-06-08 237,273 samples	0.015336	0.015963	0.015328	0.0002453	0.0002582



In this chart, a negative change means lower MAE. The zero-forecast baseline measures numerical prediction error and is not a long-short trading baseline. Pooled errors are weighted by sample counts, rather than averaging the eight fold means equally.

Sharpe is not Rank IC

Fold	Sharpe ReLU	Sharpe No ReLU	Rank IC ReLU	Rank IC No ReLU	Constant dates ReLU / No ReLU
1	0.04166	0.14194	0.01047	0.00533	2 / 0
2	0.01581	0.16344	0.00918	0.01399	6 / 0
3	-0.04511	0.03214	0.01616	-0.00781	31 / 0
4	0.01936	0.14212	0.07736	0.00117	100 / 0
5	0.12278	0.29164	0.03746	0.00854	82 / 0
6	0.10589	0.12848	0.06102	-0.00093	93 / 0
7	-0.02587	0.09257	0.01637	0.00016	63 / 0
8	0.33281	0.26228	0.01898	0.02051	32 / 0



Compare correlations on comparable dates

Native Rank IC excludes undefined correlations, so the models can have different denominators. On the **548 dates where both are defined**, the means are 0.01899 for ReLU and 0.00956 without ReLU. Across the full sample, constant-prediction dates total 409 versus 0. A constant Target can also make Rank IC undefined.

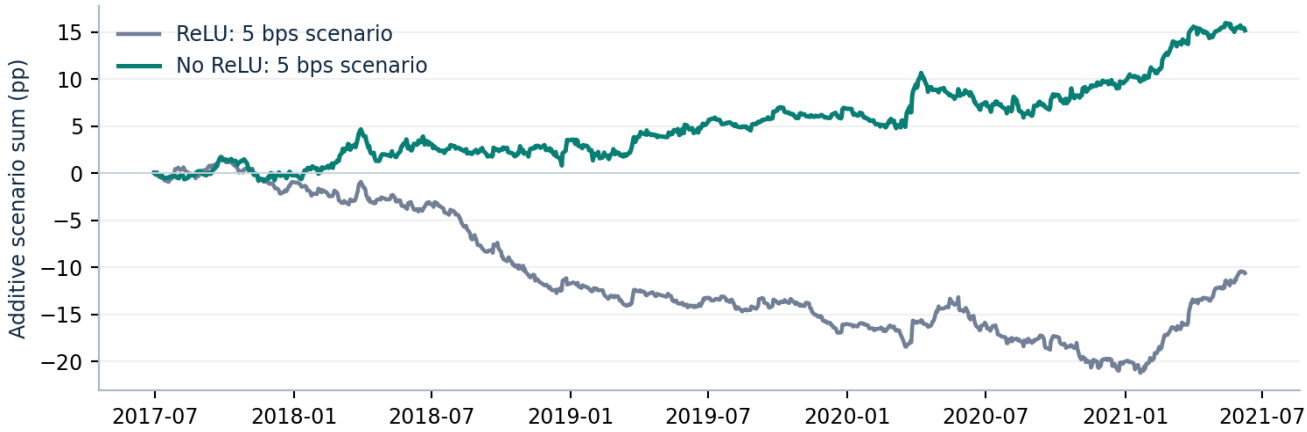
Separately labeled IID reference: daily Sharpe x sqrt(252) gives 1.219 versus 2.296. This is not adjusted for serial dependence and is not the primary metric.

TRADING COST SENSITIVITY

How much survives costs?

Illustrative cost = bps x 0.0001 x sum of absolute target-weight changes. A basis point is 0.01%. Both long and short traded notionals are charged. Start flat, carry holdings across folds and include final liquidation.

Cost / traded notional	Additive sum ReLU	Additive sum No ReLU	Sharpe ReLU	Sharpe No ReLU
0 bps	17.78 pp	35.64 pp	0.07678	0.14462
1 bps	12.10 pp	31.55 pp	0.05240	0.12831
2 bps	6.43 pp	27.46 pp	0.02786	0.11190
5 bps	-10.61 pp	15.19 pp	-0.04586	0.06219
10 bps	-39.00 pp	-5.26 pp	-0.16411	-0.02162



Turnover and break-even cost

Total absolute traded target notional is 567.77 for ReLU and 408.99 without ReLU, measured per unit of target gross exposure. The corresponding zero-additive-sum cost rates are 3.13 and 8.71 bps under this simplified model.

These are sensitivity scenarios, not realized or executable net returns. Target spans the next close to the following close. Availability of the input bar, order submission, actual fills, price drift, slippage, borrow availability/fees, taxes and liquidity constraints are not reconstructed. A proper executable backtest remains outstanding.

What the experiment can establish

Paired block bootstrap

The fixed analysis uses paired circular blocks of 5 and 20 trading dates and 2,000 resamples each. Both models use identical sampled dates. The intervals describe historical-sample sensitivity of the pooled Sharpe difference, not future profitability or selection-adjusted significance.

Block length	Replicates	95% percentile interval: No ReLU minus ReLU
5 dates	2000	[0.00018, 0.13721]
20 dates	2000	[0.00376, 0.13504]

Exclude the pilot that motivated continuation

For folds 2-8 only (836 dates), unannualized Sharpe is 0.08075 for ReLU and 0.14592 without ReLU. Gross additive sums are 16.93 pp and 32.55 pp. Common-defined Rank IC (428 dates) is 0.02138 versus 0.01077. These later folds are still not a reserved independent final test.

Earlier CNN-LSTM: contextual comparison

On 958 common dates, the prior CNN-LSTM and the no-ReLU ensemble give the following source-convention portfolio results.

Same-date measure	CNN-LSTM	No-ReLU ensemble
Unannualized Sharpe	0.09945	0.14462
Gross additive spread sum	23.60 pp	35.64 pp

This is not an architecture-only controlled test: lookback is 42 versus 21, loss is L1 versus Huber, seeds differ and stock universes may differ. The matched-row ReLU ablation is the primary comparison.

Selection and replication limits

The favorable first-fold result prompted continuation. Analysis choices were recorded before the later folds, but the study remains exploratory. There is one random seed and one historical path, no untouched final holdout, no independent replication and no adjustment for selecting this experiment after inspecting the pilot.

One model change; fixed training protocol

The removed ReLU was inside the head

Original: Linear(128,32) -> ReLU -> Linear(32,1). Variant: Linear(128,32) -> Linear(32,1). The original final Linear could already emit negative values. The frozen baseline's fold-8 diagnostic found all 32 post-ReLU features zero on 97.60% of validation rows, leaving the final bias as the forecast. Without ReLU, the two adjacent dense layers form an affine mapping at evaluation; the recurrent branches remain nonlinear.

Control	Retained setting
Branches / fusion	GRU + RNN + LSTM; hidden size 128; one layer each; mean of final hidden states; dropout 0.2
Inputs	Open, High, Low, Close, Volume, ExpectedDividend; 21 retained rows per sequence
Training	8 expanding folds x 20 epochs; batch 64; shuffle=False; float32
Objective / optimizer	HuberLoss delta=1; Adam default lr=0.001; no scheduler
State and seed	Seed 20261017; model and Adam continue across folds; no independent fold reinitializations
Runtime	Python 3.12.14; Torch 2.12.0+cpu; CPU 4 threads, interop 1

Preserved data and time ordering

Required-row validation rejects invalid prices, dates, codes, volume and Targets; duplicate security/date rows are rejected. Missing ExpectedDividend is filled with zero. Price/volume adjustment uses each security's cumulative prior adjustment factors. Scaling is fit on the training partition before purging its last two dates. Validation receives earlier per-security input context. Sequences use retained observations, which need not be consecutive calendar dates.

These source choices are preserved for an equivalent ablation, not asserted to eliminate every research bias. Filtering changes universe coverage. Scaler-before-purge uses information from the two subsequently removed dates. Cross-fold model/Adam continuation makes folds dependent. The trailing partial validation interval is excluded by the source split loop.

Portfolio convention and ties

Rank predictions within each date, average tied ranks and preserve the source sorting behavior. Long the top 200 and short the bottom 200 using linearly declining rank weights. Long weights sum to +0.5 and short weights to -0.5. Primary Sharpe is mean daily spread divided by sample standard deviation (ddof=1). This is an offline source-protocol evaluation, not a submitted Kaggle score or certification of API/rule compliance.

COMPLETION AND REPRODUCIBILITY

Verified outputs and remaining limits

Execution measure	Observed value
Completed training / validation	160 epochs; 8 folds; 8 prediction sets
Summed recorded epoch time	15.80 hours
Recorded active execution time	16.14 hours
Elapsed wall time	24.38 hours
Final model / optimizer	143,425 model parameters; 16 Adam state entries; scheduler=null

Wall time includes pauses and recovery gaps. Active time includes recorded preprocessing, training and validation; report generation and any unrecorded interrupted work are separate. No GPU migration or change to the frozen training configuration was made.

What was checked

Completion markers, all 20 epoch records in every fold, finite predictions, saved artifact hashes, exact matched validation rows/Targets/scalers, and final model tensors/normalization matching the last resumable checkpoint. The checkpoint contains model, Adam, scheduler=null, Python/NumPy/Torch random state, fold/epoch/phase, normalization, cumulative loss and previous evaluations. Initial-parameter equivalence and synthetic interrupted/resumed equivalence checks are retained in the evidence bundle.

Recovered environment

On 19 October, missing pinned packages and a truncated CSV were restored and verified before resuming from 140 epochs. A later process exit required replaying the unfinished 157th epoch from the saved model/Adam/RNG state at epoch 156. No Python traceback or cgroup OOM event was observed; the exit cause remains unknown. Abandoned partial-epoch work is excluded from recorded successful-epoch time.

Selected artifact SHA256 checks

Artifact	SHA256
Final inference weights	042cab408cde8ba7a60fa19e80301ffe f047a3b1713d767d0aae2d9de0279483
Resumable checkpoint	bb64b6035947173d8caf29509a5d36bb 5900fd65445e5a18679d7253b2aefa7a
Original CSV	bf774a86f834e5338bba74f2356ebb75 0f93a1cc4e89b064ee07251e8ed851db

Evidence: analysis/full_comparison/summary.json and paired_daily_metrics.csv; results/fold_01_metrics.json through fold_08_metrics.json; completed.json; frozen source and reference manifest; full_study_plan.json; initialization and resume-equivalence checks. The ZIP includes these sources, predictions, code and model states; the original uploaded CSV remains separate.

Conclusion boundary: historical improvements do not establish future alpha, statistical independence or a feasible net trading strategy. Multiple seeds, a reserved later period, execution-aware costs and additional ranking baselines remain unperformed follow-up experiments.